Evaluation of increment PDFs using Laplace's method

Abstract

We derive asymptotic approximations for velocity increment PDFs in the multifractal propagator framework. Using Laplace's method, we obtain two solutions: a simplified form assuming a fixed saddle point, and an exact solution via the Lambert W function. Numerical tests against the full propagator integral reveal where each approximation works best, providing groundwork for entropy calculations.

1 Introduction

Turbulent flows exhibit intermittency: velocity fluctuations become increasingly non-Gaussian at smaller scales. Understanding velocity increments $\delta_\ell u = u(x+\ell) - u(x)$ across scales ℓ is central to turbulence theory.

The multifractal approach describes intermittency through local Hölder exponents h distributed according to a spectrum $D_E(h)$. Chevillard et al. [1] proposed a propagator formalism where increment PDFs appear as integrals over the Hölder distribution, coupling Gaussian large-scale statistics with small-scale intermittency.

We use the Kolmogorov–Oboukhov lognormal model with $D_E(h) = 1 - (h - c_1)^2/(2c_2)$. This gives tractable expressions while capturing intermittency. Since the propagator integrals lack closed forms, we apply Laplace's method for asymptotic approximations at small scales.

The paper proceeds as follows. Section 2 establishes the Hölder distribution and propagator integral. Section 3 derives two approximations: assuming a fixed saddle point (Case A) and solving exactly via the Lambert W function (Case B). Section 4 introduces variance-normalized PDFs. Section 5 addresses entropy. We compare both approximations numerically against the full integral to assess their accuracy.

2 Starting point: the distribution of h

We start from (Chevillard et al., 2012, Eq. 11), which defines the probability density function of the local Hölder exponent h:

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)}}{\displaystyle\int_{h_{\min}}^{h_{\max}} \left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)} \, \mathrm{d}h}. \tag{1}$$

using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1 - \frac{(h-c_1)^2}{2c_2},$$

with $h_{\min} = -\infty$ and $h_{\max} = +\infty$, we obtain

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{(h-c_1)^2}{2c_2}\ln\left(\frac{L}{\ell}\right)\right]}{\displaystyle\int_{-\infty}^{+\infty} \exp\left[-\frac{(h-c_1)^2}{2c_2}\ln\left(\frac{L}{\ell}\right)\right] \mathrm{d}h}. \tag{2}$$

To simplify notation, we define

$$\alpha = -\ln\left(\frac{\ell}{L}\right) > 0. \tag{3}$$

Then the expression becomes

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right]}{\displaystyle\int_{-\infty}^{+\infty} \exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right] \mathrm{d}h}. \tag{4}$$

The denominator is a Gaussian integral:

$$\int_{-\infty}^{+\infty} e^{-\frac{\alpha}{2c_2}(h-c_1)^2} \mathrm{d}h = \sqrt{\frac{2\pi c_2}{\alpha}}. \tag{5}$$

Hence,

$$\boxed{\mathcal{P}_h^{(\ell)}(h) = \sqrt{\frac{\alpha}{2\pi c_2}} \exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right].} \tag{6}$$

3 Computation of the PDF using Laplace's method

3.1 General form of the PDF

We now consider the probability density function of velocity increments. According to (Chevillard et al., 2012, Eq. 17) :

$$\mathcal{P}_{\delta_\ell u}(x) = \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}} \int_{-\infty}^{+\infty} \exp[-f(h,\alpha,x)]\,\mathrm{d}h. \tag{7}$$

with

$$f(h,\alpha,x) = \alpha\left(\frac{(h-c_1)^2}{2c_2} - h\right) + \frac{1}{2}\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{8}$$

We would like to get an explicit form of our Eq. 7. To do so, we'll use Laplace's approximation (see Appendix). Knowing that

$$\forall h \in \mathbf{R}, \quad \frac{\partial f}{\partial h}(h,\alpha,x) = \alpha\left(\frac{h-c_1}{c_2} - 1\right) + \alpha\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{9}$$

we have :

$$\forall h \in \mathbf{R}, \quad \frac{\partial^2 f}{\partial h^2}(h,\alpha,x) = \frac{\alpha}{c_2} + 2\alpha^2\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \quad > 0 \tag{10}$$

We just need to verify that $h^\star$ is such that $\frac{\partial f}{\partial h}(h^\star,\alpha,x) = 0$ to be able to use Laplace approximation.

In this case, by applying the approximation to Eq. 7, we get :

$$\boxed{\mathcal{P}_{\delta_\ell u}(x) \approx \sqrt{\frac{\alpha}{2\pi c_2 \sigma^2 \frac{\partial^2 f}{\partial h^2}(h,\alpha,x))}} \exp[-f(h^\star,\alpha,x)].} \tag{11}$$

i.e

$$\boxed{\mathcal{P}_{\delta_\ell u}(x) \approx \frac{1}{\sqrt{2\pi}\sqrt{2\alpha c_2 x^2 e^{2\alpha h^\star} + \sigma^2}} \times \exp\left[-\alpha\left(\frac{(h^\star-c_1)^2}{2c_2} - h^\star\right) - \frac{1}{2}\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h^\star}\right]} \tag{12}$$

3.2 Case A: assuming $h^\star = c_1$

As a first approximation, we can consider that :

$$\frac{\partial f}{\partial h}(h^\star, \alpha, x)\Big|_{h^\star = c_1} = 0 \tag{13}$$

We can use the Laplace approximation given by Eq. 12, which becomes :

$$\boxed{\mathcal{P}_{\delta_\ell u}(x) \approx \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2\alpha c_2 x^2 e^{2\alpha c_1} + \sigma^2}} \times \exp\left[\alpha c_1 - \frac{1}{2}\left(\frac{x}{\sigma}\right)^2 e^{2\alpha c_1}\right]} \tag{14}$$

3.3 Case B: solving $\frac{\partial f}{\partial h}(h, \alpha, x) = 0$ via the Lambert W function

From (8),

$$\frac{\partial f}{\partial h}(h, \alpha, x) = \alpha\left(\frac{h - c_1}{c_2} - 1\right) + \alpha\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{15}$$

Set to zero:

$$h - (c_1 + c_2) = -c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{16}$$

Multiply by -2α and let $u = -2\alpha(h - c_1 - c_2)$:

$$u = 2\alpha c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)-u} \tag{17}$$

Consequently

$$ue^u = 2\alpha c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)} \tag{18}$$

Hence

$$\boxed{h^\star = (c_1 + c_2) - \frac{1}{2\alpha} W\left(2\alpha c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)}\right)} \tag{19}$$

where $W(x)$ is the Lambert W function (the inverse of $w \mapsto we^w$).

We can now plug the obtained $h^\star$ into Eq. 12.

4 Normalized PDF

According to (Chevillard et al., 2012, Eq. 12), the p-th order moment of the velocity increment is expressed as

$$\langle|\delta_\ell u|^p\rangle \underset{\ell\to 0}{\approx} \sigma^2 \langle|\delta|^p\rangle \left(\frac{\ell}{L}\right)^{\zeta_p}, \quad \text{with} \quad \zeta_p = \min_h\big[ph + 1 - \mathcal{D}^E(h)\big] \quad \text{and} \quad \delta \sim \mathcal{N}(0,1). \tag{20}$$

Using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1 - \frac{(h - c_1)^2}{2c_2}, \tag{21}$$

the minimization problem is

$$\frac{\mathrm{d}}{\mathrm{d}h}\big[ph + 1 - \mathcal{D}^E(h)\big] = 0 \quad i.e. \quad \frac{\mathrm{d}}{\mathrm{d}h}\big[ph + \frac{(h - c_1)^2}{2c_2}\big] = 0 \tag{22}$$

Hence, the minimizing exponent is

$$h_p = c_1 - pc_2, \tag{23}$$

and inserting this value into Eq. 20 gives

$$\zeta_p = pc_1 - \frac{p^2}{2}c_2 \tag{24}$$

In particular, for the second-order :

$$\zeta_2 = 2(c_1 - c_2) \tag{25}$$

Knowing that $\langle|\delta|^2\rangle = 1$, we get :

$$\boxed{\langle(\delta_\ell u)^2\rangle \underset{\ell\to 0}{\approx} \sigma^2 \left(\frac{\ell}{L}\right)^{2(c_1-c_2)}.} \tag{26}$$

We now define a normalized increment

$$\boxed{\delta_\ell \nu = \frac{\delta_\ell u}{\sqrt{\langle(\delta_\ell u)^2\rangle}} = \frac{\delta_\ell u}{\sigma\left(\frac{\ell}{L}\right)^{c_1-c_2}}} \tag{27}$$

This normalization ensures that

$$\langle(\delta_\ell \nu)^2\rangle = 1, \tag{28}$$

for all scales ℓ/L, so that all normalized PDFs can be directly compared across scales.

We thus have $\delta_\ell \nu = \delta_\ell u/\sigma\left(\frac{\ell}{L}\right)^{c_1-c_2} = g(\delta_\ell u)$ where $g(x) = x/\sigma\left(\frac{\ell}{L}\right)^{c_1-c_2}$ is monoton and differentiable.

Knowing that : $\mathcal{P}_{\delta_\ell \nu}(x) = \left|\frac{1}{g'(g^{-1}(x))}\right| \mathcal{P}_{\delta_\ell u}(g^{-1}(x))$, we have : $\mathcal{P}_{\delta_\ell \nu}(x) = \sigma\left(\frac{\ell}{L}\right)^{c_1-c_2} \mathcal{P}_{\delta_\ell u}(\sigma\left(\frac{\ell}{L}\right)^{c_1-c_2} x)$ i.e.

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(x) = \sigma e^{\alpha(c_2-c_1)} \mathcal{P}_{\delta_\ell u}(\sigma e^{\alpha(c_2-c_1)} x)} \tag{29}$$

We then obtain from Eq. 12 the normalized form of the PDF :

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(x) \approx \frac{e^{\alpha(c_2-c_1)}}{\sqrt{2\pi}\sqrt{2\alpha c_2 x^2 e^{2\alpha(h^\star+c_2-c_1)}+1}} \times \exp\left[-\alpha\left(\frac{(h^\star-c_1)^2}{2c_2} - h^\star\right) - \frac{1}{2}x^2 e^{2\alpha(h^\star+c_2-c_1)}\right]} \tag{30}$$

which is valid if $h^\star$ is such that $\frac{\partial f}{\partial h}(h^\star, \alpha, \sigma e^{\alpha(c_2-c_1)}x) = 0$.

4.1 Case A: assuming $h^\star = c_1$

If we consider that $\frac{\partial f}{\partial h}(h^\star, \alpha, \sigma e^{\alpha(c_2-c_1)}x)\Big|_{h^\star=c_1} = 0$, then we get from Eq. 30 :

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(x) \approx \frac{1}{\sqrt{2\pi}\sqrt{2\alpha c_2\, x^2 e^{2\alpha c_2}+1}} \times \exp\left[\alpha c_2 - \frac{1}{2}x^2 e^{2\alpha c_2}\right]} \tag{31}$$

4.2 Case B: solving $\frac{\partial f}{\partial h}(h, \alpha, \sigma e^{\alpha(c_2-c_1)}x) = 0$ via the Lambert W function

We want to find $h^\star$ such that $\frac{\partial f}{\partial h}(h^\star, \alpha, \sigma e^{\alpha(c_2-c_1)}x) = 0$. In Eq. 19, by considering that $h^\star$ is a function of x, we just need to replace x by $\sigma e^{\alpha(c_2-c_1)}x$. We get :

$$\boxed{h^\star = (c_1+c_2) - \frac{1}{2\alpha}W\big(2\alpha c_2 x^2 e^{4\alpha c_2}\big)} \tag{32}$$

We can now plug the obtained $h^\star$ into Eq. 30. Let $W = W\big(2\alpha c_2 x^2 e^{4\alpha c_2}\big)$. We get :

$$\boxed{\mathcal{P}_{\delta_\ell\nu}(x) \approx \frac{1}{\sqrt{2\pi}\sqrt{2\alpha c_2 x^2 e^{4\alpha c_2 - W} + 1}} \times \exp\left[-\frac{\alpha}{2}\left(\frac{1}{c_2}\left(\frac{W}{2\alpha}\right)^2 - 3c_2\right) - \frac{1}{2}x^2 e^{4\alpha c_2 - W}\right]} \tag{33}$$

i.e.

$$\boxed{\mathcal{P}_{\delta_\ell\nu}(x) \approx \frac{e^{\frac{3}{2}} c_2 \alpha}{\sqrt{2\pi}} \frac{e^{-\frac{2W+W^2}{8c_2\alpha}}}{\sqrt{1+W}}} \tag{34}$$

5 Entropy of the normalized PDF

We now compute the Shannon entropy of the normalized increment PDF $\mathcal{P}_{\delta_\ell\nu}$ obtained in Eq. 28. By definition,

$$S(\delta_\ell\nu) = -\int_{-\infty}^{+\infty} \mathcal{P}_{\delta_\ell\nu}(x)\, \ln\big[\mathcal{P}_{\delta_\ell\nu}(x)\big]\, \mathrm{d}x \tag{35}$$

References

[1] L. Chevillard, B. Castaing, A. Arneodo, E. Lévêque, J.-F. Pinton, and S. G. Roux, *A phenomenological theory of Eulerian and Lagrangian velocity fluctuations in turbulent flows*, C. R. Physique **13**, 899–928 (2012).

A Appendix - Laplace's approximation theorem

Let $I = [a, b]$ be an interval, $\varphi \in C^2(I, \mathbb{R})$ and $g \in C^0(I, \mathbb{C})$. Assume that:

(i) For all $t > 0$, $\displaystyle\int_a^b e^{t\varphi(x)}|g(x)|\, dx < +\infty$;

(ii) φ' vanishes at a unique point $x_0 \in I$ and $\varphi''(x_0) < 0$;

(iii) $g(x_0) \neq 0$.

Then

$$\int_a^b e^{\,t\varphi(x)}\, g(x)\, dx \sim \sqrt{\frac{2\pi}{t\,|\varphi''(x_0)|}}\, e^{\,t\varphi(x_0)}\, g(x_0) \qquad \text{as } t \to +\infty.$$